E

NAME ____________________

Graphing and Tables

Graphing Rules
- Always include a title
- Give headings to each axis
- Independent Variable is on the x-axis
- Dependent Variable is on the y-axis
- Scale should be even

Table Rules
- Always include a title
- Give headings to all the columns
- The left hand column is usually the Independent Variable

Investigation I
A scientist put nine groups of silver perch eggs in water at nine different temperatures and observed how long it took for the eggs to hatch at the different temperatures.

Silver Perch Hatchings

Water Temperature (°C)	**Hatching Time (Days)**
1	42
3	23
4	20.5
5	17
6	15
8	12.5
10	10
12	10
14	8.5

1) What was the independent variable?____________________

2) What was the dependent variable? ____________________

3) What hypothesis was being tested?

If the ____________________ increases the ____________________ for silver
independent variable dependent variable
perch eggs decrease.

4) What type of graph would you use to graph the data? (Line or Column)

____

6) As the water temperature increased did the hatching time decrease? ____________________

What temperature would you use if you wanted to hatch Silver Perch quickly? ____________________

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Investigation II

A chemist was investigating the effect of water temperature on how fast sugar dissolves. He collected the following data.

At 45°C sugar dissolves in 16 seconds
At 30°C sugar dissolved in 22 seconds
At 25°C sugar dissolved in 24 seconds
At 40°C sugar dissolved in 18 seconds
At 35°C sugar dissolved in 20 seconds

1) Put the data into the table below and insert the units in the brackets.

Temperature ( )	Dissolving time ( )
25°C	
45°C	

2) What was the independent variable? ______________________________

3) What was the dependent variable? ______________________________

4) What was the hypothesis being tested?

If the ______________________ increases the ______________________
independent variable   dependent variable
decreases.

Investigation III

A Teacher wanted to know what the most popular colour that her students liked. She collected the below data from her students.

Colour popularity of students

Colour	**Number of Students who like the colour**
Blue	7
Green	5
Purple	4
Red	9

1) What was the independent variable? ______________________________

2) What was the dependent variable? ______________________________

3) What was the hypothesis? ______________________________

____

4) Graph the data using a Column graph on the paper provided.

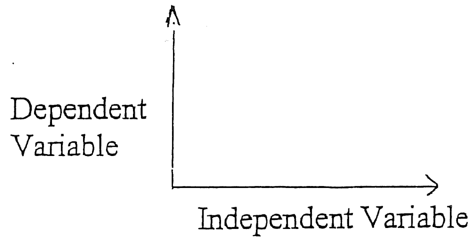
Independent and Dependent Variables

A **variable** is anything that can affect the result of an experiment.

An **Independent Variable** (sometimes called an experimental variable) is the thing the scientist will deliberately change during an experiment. It is what the scientist is testing. This goes in the first column of a table and on the bottom of a graph.

The **Dependent Variable** is the thing the scientist will measure to get the results. They do the experiment to get these results. This goes in the second column of a table and up the side of a graph.

Independent Variable	Dependent Variable



In the following scientific questions underline the Independent variable and draw a circle around the dependent variable in each question or statement.
(example) Does the colour of a person's hair affect their intelligence?

1. Does the colour of a car affect how fast it can go?
2. Is the rate of tooth decay affected by the type of toothpaste you use?
3. Does a cup of coffee made with milk or water cool down the quickest?
4. Does adding sugar make a cup of coffee cool down quicker?
5. Is the rate of cooling of a cup of coffee affected by how much I stir it?
6. Does the amount of protein in a child's diet affect their rate of growth?
7. Is the temperature inside a car affected by the colour of the car?
8. Will a sponge cake mix rise more if the oven temperature is increased?
9. What is the favourite C.D. amongst year 8 students?
10. Do students do better at school if they eat breakfast?
11. Does doing more homework improve your marks at school?
12. Is the crop yield in a paddock improved by adding more fertilizer?
13. The more often you brush your teeth the less cavities you develop.
14. If an athlete takes more steroids will they become stronger?
15. How tall you will be depends on how tall your parents are?
16. The narrower a bike tyre is the faster the bike will go.
17. The wider the bike tyre the longer the breaking distance needed to stop.
18. Different ways of kicking will make a football go different distances.
19. Some ways of kicking a football are more accurate than others.
20. Left handed people get higher marks than right handers at school.

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For each item below, specify the independent and dependent variables, as well as constants.

1. A study was done to find if different tire treads affect the braking distance of a car.

I: ______________ **D:** ________________ **C:** ______________

2. The time it takes to run a mile depends on the person's running speed.

I: ______________ **D:** ________________ **C:** ______________

3. The height of bean plants depends on the amount of water they receive.

I: ______________ **D:** ________________ **C:** ______________

4. The higher the temperature of the air in the oven, the faster a cake will bake.

I: ______________ **D:** ________________ **C:** ______________

5. Lemon trees receiving the most water produced the most lemons.

I: ______________ **D:** ________________ **C:** ______________

6. An investigation found that more bushels of potatoes were produced when the soil was fertilized more.

I: ______________ **D:** ________________ **C:** ______________

7. Students measured the temperature of the water at different depths in Lake Skywalker and found that the temperature varied.

I: ______________ **D:** ________________ **C:** ______________

8. The amount of pollution produced by cars was measured for cars using gasoline containing different amounts of lead.

I: ______________ **D:** ________________ **C:** ______________

9. Four groups of rats are first massed and then fed identical diets except for the amount of vitamin A they receive. Each group gets a different amount. After 3 weeks on the diet, the rats' masses are measured again to see if there has been a decrease.

I: ______________ **D:** ________________ **C:** ______________